

10 min of simulation of a persisting wake

Inflow

Wind turbine
1 & 2

Wake
1 & 2

Instantaneous 2D slices

$(t_{1...n}, x_{-2D}, y[-2D:2D], z[0:2000m])$
 $\rightarrow u, v, w, \theta$

Temporally averaged 2D slices

$(x_{-2D}, y[-2D:2D], z[0:2000m])$
 $\rightarrow \bar{u}, \bar{v}, \bar{w}, \bar{\theta}, \overline{TKE_{SGS}},$
 $\overline{u'u'}, \overline{v'v'}, \overline{w'w'}, \overline{u'v'}, \overline{v'w'}, \overline{u'w'}, \overline{w'\theta'}$

Time series

$(t_{1...n})$
 $\rightarrow$ Momentum/heat flux at the surface

Radial profiles

$(\text{sector } 4^\circ, \text{ radial pos } 2m)$

$\rightarrow \bar{\alpha}, \bar{c}_l, \bar{c}_d$

$\rightarrow \overline{dF_n/dr}, \overline{dF_t/dr}$ (chord normal and tangential force per unit length)

$\rightarrow$ Thrust $\bar{T}$ and torque $\bar{Q}$

$\rightarrow$ 10-min min, max and σ

Instantaneous 2D slices

$(t_{1...n}, x[:,0.5D], y[-2D:2D], z[0:300m])$
 $\rightarrow u, v, w, \theta$

Temporally averaged 2D slices

$(x[:,0.5D], y[-2D:2D], z[0:300m])$
 $\rightarrow \bar{u}, \bar{v}, \bar{w}, \bar{\theta},$
 $\overline{u'u'}, \overline{v'v'}, \overline{w'w'}, \overline{u'v'}, \overline{v'w'}, \overline{u'w'}, \overline{w'\theta'}$